SIGNSPEAK: Integrating Computer Vision and Speech Synthesis for Alphabet Filipino Sign Language (FSL)

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***Abstract*—In our interconnected world, effective communication is a fundamental human right fostering social interaction, education, and personal growth. However, individuals with sensory impairments, such as hearing loss, encounter significant barriers in communication, hindering their participation in society. This challenge is pronounced in the Philippines, where Filipino Sign Language (FSL) is vital for the Deaf community. The "SignSpeak" study addresses these challenges by integrating computer vision and speech synthesis technologies. Utilizing platforms like Raspberry Pi and Jetson Nano, SignSpeak interprets hand gestures in real-time, converting them into audible speech, and translates spoken language into text, enabling bidirectional communication. This approach empowers individuals with varying sensory abilities, fostering inclusivity and enhancing educational opportunities. SignSpeak promotes social equity and empowers the Deaf community in the Philippines through rigorous software and hardware development and a commitment to linguistic inclusivity. Beyond technical innovations, SignSpeak envisions a future where individuals of all abilities can communicate freely and participate in society. The study results were highly successful, demonstrating 95% accuracy, 96% precision, 94% recall, and a 95% F1 score. Additionally, the prototype received positive feedback, with an overall mean score of 4.845. These outcomes highlight SignSpeak's effectiveness and potential impact in improving communication accessibility for the Deaf community in the Philippines.**

***Keywords*—Filipino Sign Language, Deaf Community, Communication Accessibility, Computer Vision, Speech, Synthesis, Real-time Interpretation, Inclusivity, SignSpeak study, Technology Integration, Social Equity**

I. INTRODUCTION

In the rapidly evolving digital era, the importance of inclusive communication cannot be overstated. For individuals with sensory impairments, such as hearing loss, communication barriers persist due to the limited availability of accessible communication tools. The ability to communicate effectively is not just a fundamental human right but also a cornerstone of social interaction, education, and personal development [1]. However, for a significant portion of the global population, communication barriers persist due to sensory impairments, such as hearing loss, and the limited availability of accessible communication tools [2].

In the Philippines, a nation known for its rich cultural diversity, there exists a vibrant Deaf community. According to the 2000 Philippine census, an estimated 121,000 Deaf Filipinos were identified. Despite this sizable population, the prevalence of sign language users in the country remains relatively low. This disconnect between the Deaf community and mainstream society underscores the pressing need for innovative solutions to bridge the communication gap and promote inclusiveness. Although there is a significant Deaf population, a noticeable gap remains in effective communication support. Even with the importance of sign language, many Deaf Filipinos still face communication barriers in school, at work, and in other areas of their lives. There is a shortage of FSL interpreters, and FSL is not as widely known or utilized in the Philippines as it should be. However, there is a growing awareness of the significance of FSL in the Philippines, and many organizations are working to promote FSL and provide services to the Deaf community.

The "SignSpeak" study addresses this issue by integrating computer vision and speech synthesis technologies. This innovative approach aims to facilitate real-time interpretation of hand gestures into audible speech and convert spoken language into text, enabling bidirectional communication. SignSpeak has the potential to revolutionize how individuals with sensory impairments interact with the world, promoting greater accessibility and inclusivity in various aspects of daily life. The general objective of the study is to design and develop an integrated communication device for seamless interaction between individuals with auditory and visual impairments. This device aims to provide real-time translation of hand gestures into audible speech and convert spoken language into text. Specific objectives include developing a hardware microcomputer for the SignSpeak system, integrating computer vision algorithms for hand gesture recognition and speech synthesis for real-time interpretation; creating software to provide an inclusive tool for effective communication for deaf individuals in diverse environments; and innovating the system to support bidirectional communication, allowing users to input signs and voice for translation and receive real-time synthesized speech. The scope includes hardware and software development within a limited timeframe, focusing on inclusivity and accessibility. The system's design considers public health and safety, cultural adaptation, societal benefits, and environmental considerations. By achieving these objectives, the study aims to contribute to the field of computer engineering and improve sign language interpretation. Fig. 1 presents the conceptual framework of the system.

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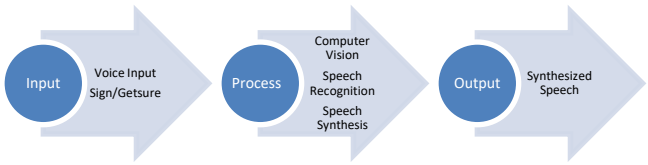


Fig. 1. Conceptual Framework

Fig. 1 depicts the conceptual framework guiding the development of 'SignSpeak,' aiming to create an accessible system bridging sign language and spoken language. Utilizing the Input-Process-Output (IPO) model, the study outlines clear inputs—voice and visual sign language gestures captured via microphone and camera, respectively. These inputs undergo processing: computer vision interprets gestures, while speech recognition converts voice to text. Integrated outputs include synthesized speech, facilitating seamless communication by transforming diverse inputs into coherent speech output, enhancing accessibility across user needs

II. METHODOLOGY

The study employed a Rapid Application Development (RAD) software development lifecycle for the planning and implementation of the device. Rapid Application Development (RAD). RAD places a high emphasis on swift prototyping, real-time feedback, and ongoing adaptation, favoring these over prolonged planning and development phases. This approach offers the advantage of flexibility and adaptability during the creative process, enabling continuous input and evolution as the project unfolds. By reducing the planning stage and maximizing prototype development, the study aims to effectively tackle time constraints and maintain a consistent pace of progress [6]. Fig. 2 shows the hardware representation of the system.

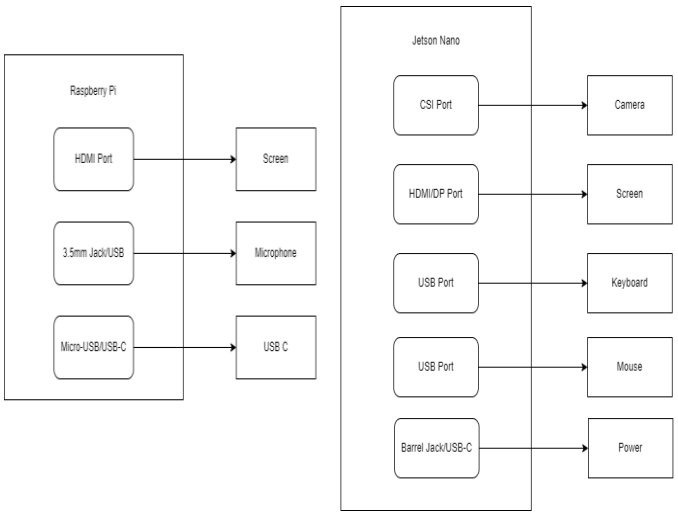


Fig. 2. Hardware Representation of the System

As illustrated in Fig. 2, the hardware connections demonstrate how these components are integrated to achieve the desired functionalities. The first step in hardware development was to identify and select the appropriate components that would support the functionalities required by the system. Complementing this setup, the NVIDIA Jetson Nano managed intensive computational tasks with its USB connectivity for input devices, HDMI/DisplayPort for screen output, CSI port for camera integration, and power via barrel jack or USB-C. The configured settings across both platforms enabled seamless communication and data transfer critical for real-time gesture recognition and interpretation.

The software development began with data gathering, where detailed requirements were collected to ensure the system met the needs of Deaf individuals, sign language interpreters, and other stakeholders. This phase included a comprehensive review of existing technologies in computer vision, speech synthesis, and sign language interpretation to identify gaps and opportunities. Engaging with members of the Deaf community, sign language interpreters, and educators helped understand their specific needs, challenges, and preferences, which were documented along with the functional and non-functional requirements, emphasizing real-time processing, high accuracy in gesture recognition, and user-friendly interfaces. A dataset of FSL alphabet gestures and corresponding voice samples was prepared for training and testing the system.

Based on these requirements, the system design phase involved creating seamless system architecture and selecting components such as the Raspberry Pi for processing, NVIDIA Jetson Nano for enhanced GPU capabilities, and appropriate cameras and microphones. Suitable computer vision algorithms like Mediapipe and TensorFlow were chosen for hand gesture recognition, alongside text-to-speech algorithms for speech synthesis. The system architecture developed included modules for gesture recognition, speech synthesis, and the user interface.

The software development covers the following: (a) Computer Vision Module for the implementation of gesture recognition using Python libraries such as OpenCV and Mediapipe, ensuring accurate detection and interpretation of FSL gestures; (b) Speech Synthesis Module for the development of the speech synthesis component using text-to-speech libraries, converting recognized gestures into audible speech; (c) Integration for the computer vision and speech synthesis modules to work cohesively, allowing seamless translation of gestures to speech and vice versa; (d) User Interface was designed a user-friendly interface using Tkinter for ease of interaction, enabling users to input and receive translations.

The software development process for the 'SignSpeak' project involved several key steps to ensure the system could accurately interpret Filipino Sign Language (FSL) and hand gestures. This includes collecting data, creating datasets, training machine learning models, and developing a user-friendly interface. Figs. 3 and 4 illustrate the DFD levels and their interaction with the software and hardware components of the design project.

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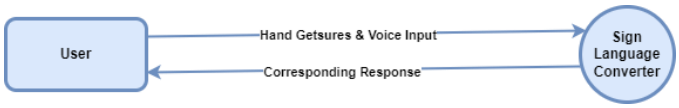


Fig. 3. DFD Level 1

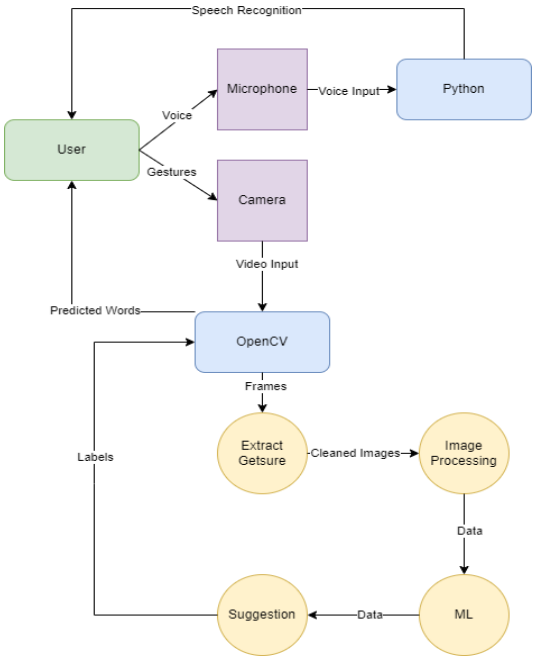


Fig. 4. DFD Level 2

Fig. 5 illustrates the sequential flow of modules involved in algorithm development, detailing the path from data collection through preprocessing, feature extraction, model training, and finally to the output stage. The Algorithm development focused on creating efficient and accurate methods to detect and recognize hand gestures from video streams, and in the study, Data Collection, Data Preprocessing, Feature Extraction, and Data Modeling were used.

The integration of the SignSpeak system involves connecting hardware components such as Raspberry Pi and Jetson Nano to ensure seamless operation and effective communication between devices, as shown in Figs. 6 and 7. Raspberry Pi facilitates user interaction through its HDMI port for display connectivity, 3.5 mm audio jack or USB port for microphone input, and additional USB ports for peripherals as required. It receives power via micro-USB or USB-C connections. In contrast, Jetson Nano handles computational tasks, utilizing USB ports for keyboard and mouse connectivity, HDMI or DisplayPort for display output, CSI port for camera input, and either barrel jack or USB-C for power supply. Both systems were configured to maintain stable connectivity, allowing for efficient software updates and troubleshooting. This integration supports the system's functionalities, enabling real-time recognition and interpretation of Filipino Sign Language (FSL) gestures through synchronized hardware interactions and data exchange between the Raspberry Pi and Jetson Nano platforms.

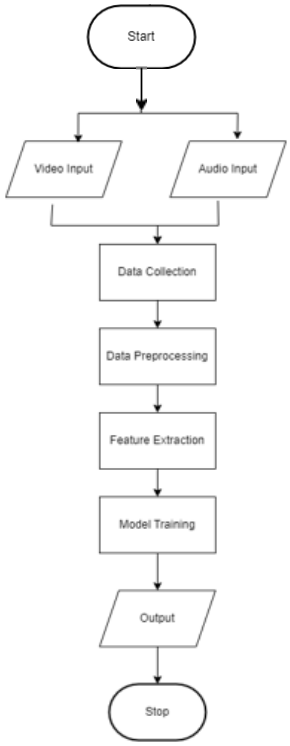


Fig. 5. Algorithm flowchart

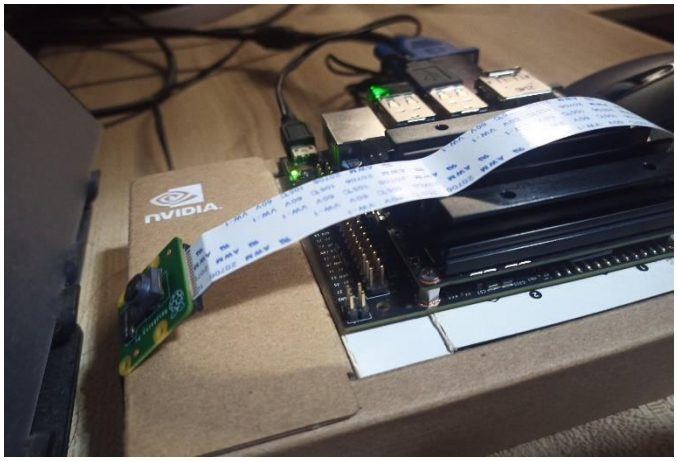


Fig. 6. Actual Prototype

The performance of the SignSpeak system was rigorously evaluated using accuracy metrics calculated with the aid of scikit-learn, pandas, and numpy libraries. These metrics provided comprehensive insights into the system's ability to classify FSL gestures accurately.

The dataset was meticulously divided into training and testing sets using a stratified random sampling technique to ensure a balanced distribution of samples across classes. Subsequently, the RandomForest Classifier model was trained on the training set and evaluated on the testing set.

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Fig. 7. Jetson Nano Setup

The formula for Accuracy Metrics is as follows:

Accuracy. The proportion of correctly classified samples to the total number of samples.

$$Accuracy\ (ACC) = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

Precision. The ratio of correctly predicted positive observations to the total predicted positive observations.

$$Precision\ = \frac{TP}{TP+FP} \quad (2)$$

Recall. The ratio of correctly predicted positive observations to all observations in the actual class.

$$Recall\ = \frac{TP}{TP+FN} \quad (3)$$

F1 Score. The weighted average of precision and recall.

$$F1\ Score\ = 2 * \frac{Precision*Recall}{Precision+Recall} \quad (4)$$

III. Results

The SignSpeak system demonstrated an impressive accuracy of 95%, signifying its ability to correctly classify 95% of the test samples. Furthermore, the high precision, recall, and F1 score underscore the effectiveness of the system in accurately recognizing FSL gestures. These accuracy metrics offer valuable insights into the performance of the SignSpeak system and underscore its potential for real-world applications in facilitating communication for individuals using FSL as shown in Table 1.

Table 1. Summary of the Results of Evaluation

Metric	Score
Accuracy	95%
Precision	96%
Recall	94%
F1Score	95%

Fig. 8 illustrates the detected hand's landmarks, which are then normalized and stored as feature vectors alongside corresponding class labels, facilitating the subsequent training of the classification model. The processed data is serialized into a pickle file format, preserving its structure and enabling seamless integration into machine learning pipelines.

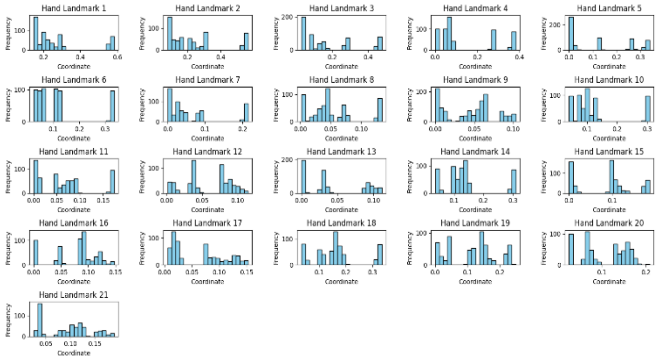


Fig. 8. Hand Landmark Graph

The confusion matrix in Fig. 9 illustrates the performance of the SignSpeak system in classifying Filipino Sign Language (FSL) gestures.

Each row represents the actual class, while each column represents the predicted class. The diagonal elements indicate the number of correctly classified samples for each class, while off-diagonal elements represent misclassifications. Table 2 shows the results of the SL Interpreter questionnaire.

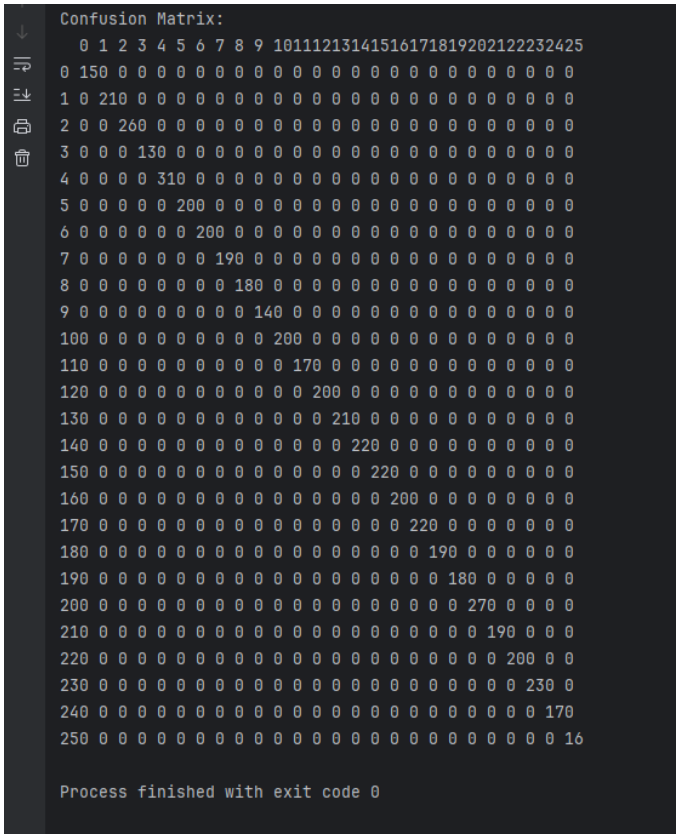


Fig. 9. Matrix Result

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The model was trained using the training set for a specified number of epochs, with validation performed after each epoch to monitor the model's performance. Finally, the model is evaluated on both the validation and test sets to assess its accuracy. This iterative approach allows for fine-tuning the model and ensuring its robustness across different datasets.

The User Acceptance and Functionality Testing was conducted with 3 SL Interpreters and 40 students for a total of 43 respondents and the results are stated in Tables 2 and 3.

Table 2. *SL Interpreters' questionnaire results*

	1	2	3	4	5	Mean Scores
How familiar are you with FSL?	0	0	0	0	3	5
How accurately was the sign language translated?	0	0	0	2	1	4.3
How accurately does your voice translate into text?	0	0	0	0	3	5
Rate the accuracy of the custom hand gesture you created	0	0	0	0	3	5
Overall, how would you rate your impression of SignSpeak?	0	0	0	0	3	5
How likely are you to use systems like SignSpeak?	0	0	0	1	2	4.6
Overall Mean Score from the prototype = 4.78						

Table 3. *Student Questionnaire Results*

	1	2	3	4	5	Mean Scores
How familiar are you with FSL?	22	9	9	10	0	2.14
How accurately was the sign language translated?	0	0	0	2	38	4.95
How accurately does your voice translated into text?	0	0	0	0	40	5
Rate the accuracy of the custom hand gesture you created	0	0	0	9	31	4.775
Overall, how would you rate your impression of SignSpeak?	0	0	1	3	36	4.875
How likely are you to use a system like SignSpeak?	0	0	0	2	38	4.95
Overall Mean Score from the prototype = 4.91						

IV. Conclusion

In conclusion, the 'SignSpeak' study represents a significant advancement in overcoming communication barriers for the deaf and hard-of-hearing community in the Philippines. By integrating computer vision and speech synthesis technologies, the system facilitates real-time interpretation of Filipino Sign Language (FSL) and hand gestures, enabling seamless bidirectional communication. Beyond technical innovation, 'SignSpeak' promotes inclusivity, cultural awareness, and educational empowerment among deaf individuals. Its scalability and sustainability underscore its potential to foster a more inclusive society where everyone can communicate effectively and participate fully in societal activities, showcasing technology's transformative impact on social equity and accessibility.

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